

Mount Kenya University

Unlocking Infinite Possibilities

Biao Wang School Management System

Unit Code: BIT3208

Unit Name: Advanced Web Design and Development

Student Name: Kagochi Brian Kahora

Admission Number: BSCCS/2024/54776

Course: Computer science

Lecturer Name: Michael Nyoro

Technologies Used:

HTML, CSS, PHP, MySQL, XAMPP

GitHub/Portfolio Link: <https://github.com/kahosh-dev/BIT-3208-ADVANCED-WEB-DESIGN-AND-DEVELOPMENT>

Table of Contents

Table of Contents

WEEK 1: Local Environment Setup.....	3
• Installation and Testing of Local Development Environment	
• Required Evidence	
• Student Reflection	
WEEK 2: Wireframes and GUI Design	6
• User Interface Planning and System Design	
• Required Evidence	
• Student Reflection	
WEEK 3: JavaScript and PHP Basics	9
• Frontend Interaction and Backend Foundations	
• Required Evidence	
• Student Reflection	
WEEK 4: Backend Development and Authentication.....	12
• Backend Processing and Authentication Systems	
• Modern Backend Alternatives	
• Student Reflection	
WEEK 5: Database Integration and CRUD Operations	15
• Database-Driven Application Development	
• Creating Databases	
• Creating Tables	
• CRUD Operations	
• SQL Database Table Creation Script	
• Student Reflection	
References.....	18

WEEK 1: Local Environment Setup

Installation and Testing of Local Development Environment

Required Evidence

Fig 1: Installation of XAMPP on Windows

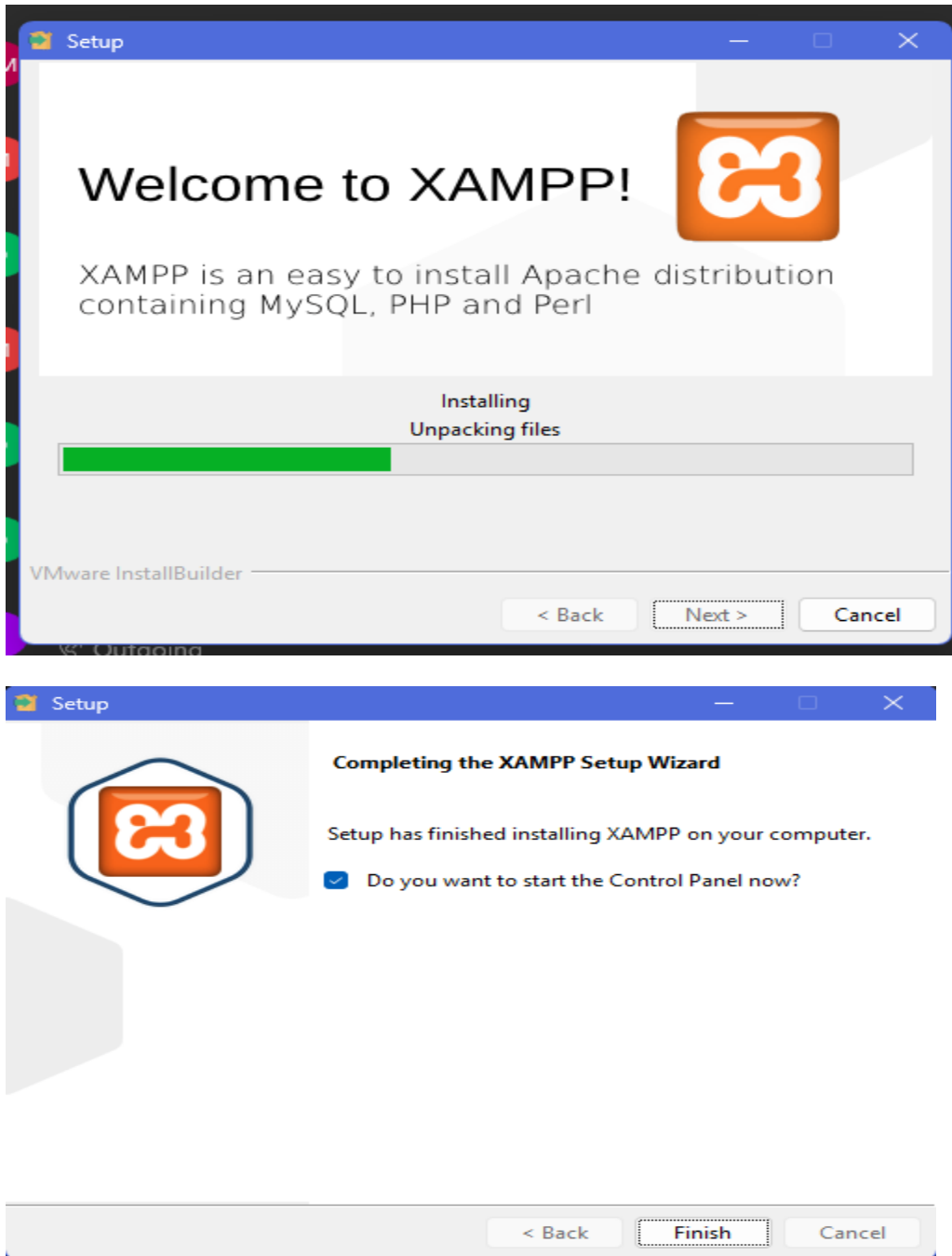


Fig 2: Apache and MySQL services running in XAMPP Control Panel

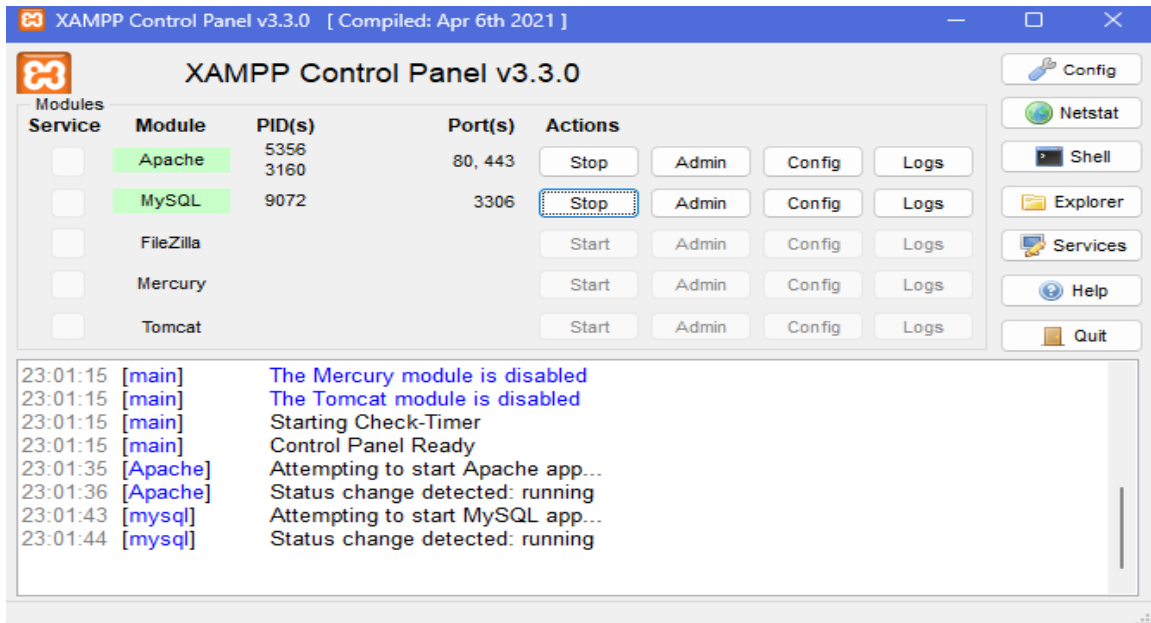


Fig 3: Localhost test page loading successfully in browser

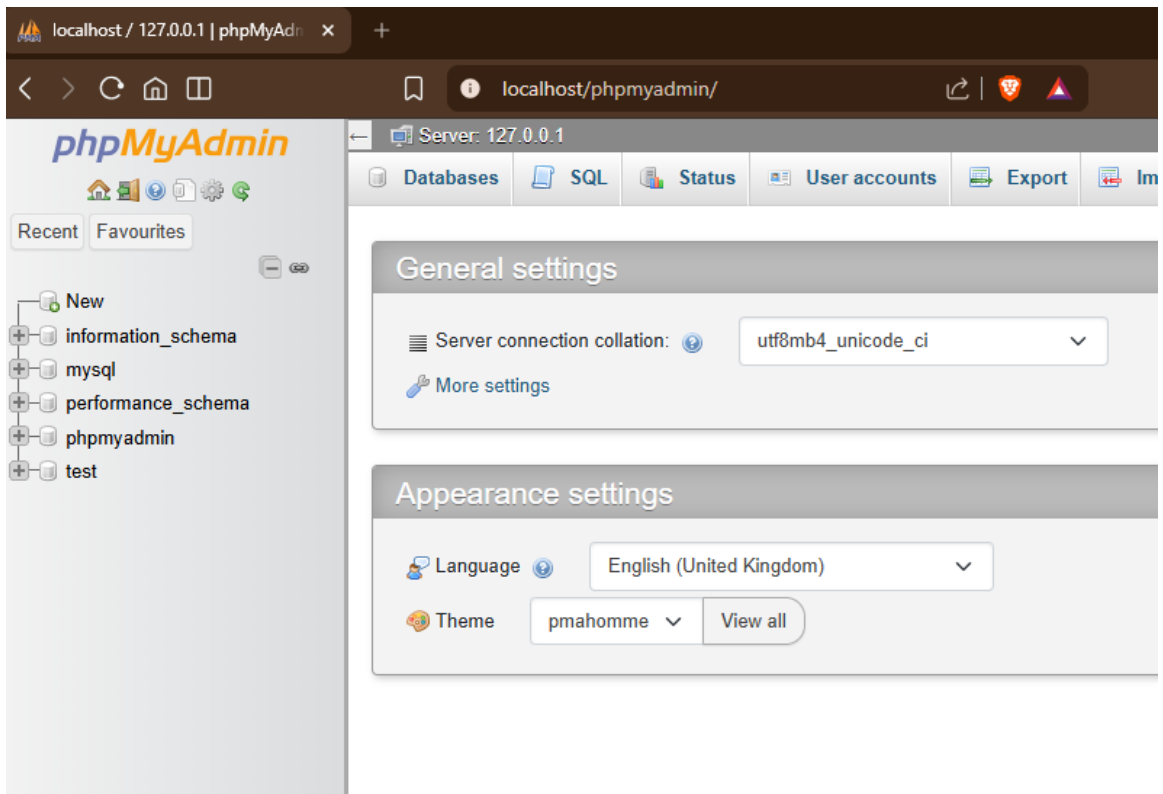


Fig 4: First PHP Hello World test page rendered in browser

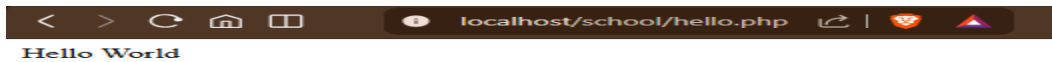


Fig 5: Successful database connection test to school_system_db using PHP mysqli_connect ()

```
db.php > ...
1  <?php
2
3  /* DATABASE CONFIGURATION */
4
5  $DB_HOST = "localhost";
6  $DB_USER = "root";
7  $DB_PASS = "";
8  $DB_NAME = "school_system";
9
10 /* CREATE CONNECTION */
11
12 $mysqli = new mysqli(
13     $DB_HOST,
14     $DB_USER,
15     $DB_PASS,
16     $DB_NAME
17 );
18
19 /* CHECK CONNECTION */
20
21 if($mysqli->connect_error){
22     die(
23         "Database Connection Failed: "
24         . $mysqli->connect_error
25     );
26 }
27
28 /* SET CHARACTER */
29
30 $mysqli->set_charset("utf8mb4");
31
32 ?>
```

STUDENT REFLECTION

Week 1 I learned how to set up a local development environment XAMPP and Apache and MySQL services. I was able to successfully test localhost functionality, create my first PHP Hello World page, and connect PHP to MySQL databases. This week gave me a better understanding of server-side web development and localhost environments.

WEEK 2: Wireframes and GUI Design
User Interface Planning and System Design

Required Evidence

Fig 1: Hand-drawn wireframe of the Login page

A hand-drawn wireframe of a login page. At the top center is a blue arrow icon pointing right, followed by the word "Login" in blue. Below this are two light gray rectangular input fields. The first field is labeled "Username" and the second is labeled "Password". Below the password field is a small gray square representing a checkbox. Below the checkbox is the text "Remember Me". At the bottom is a wide blue rectangular button with a white lock icon and the word "Login" in white. To the left of the button, below the "Remember Me" text, is a blue arrow pointing left followed by the word "Back" in blue.

Fig 2: Dashboard layout design showing KPI cards and navigation

A dashboard layout design for a "School Management System". The top header is a solid blue bar containing a circular logo on the left with the letters "SMS" and the text "School Management System" in white on the right. Below the header, the main content area has a white background. It starts with the text "Welcome to Student Marks Management System" in bold, followed by "Register or Login As Student or Admin". Below this are two white rectangular boxes with rounded corners. The left box is titled "Student Portal" with a graduation cap icon. It contains two blue buttons: "Student Registration" and "Student Login". The right box is titled "Admin Portal" with a person icon. It contains two blue buttons: "Admin Registration" and "Admin Login".

Fig 3: Mobile view mockup

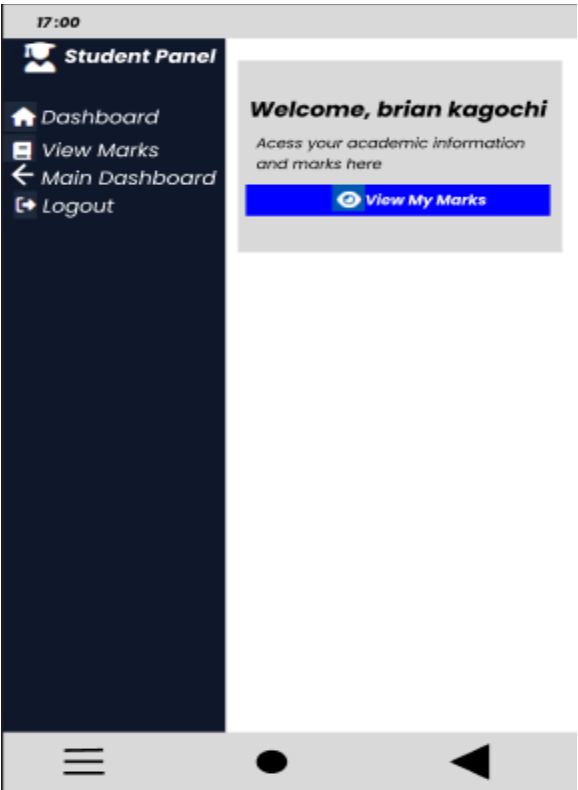


Fig 4: Navigation Structure Plan for the School Management System.

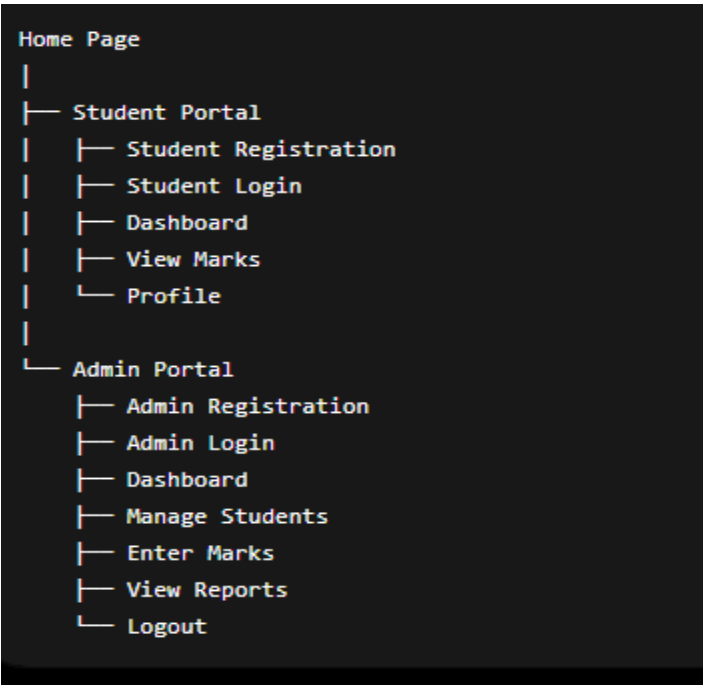
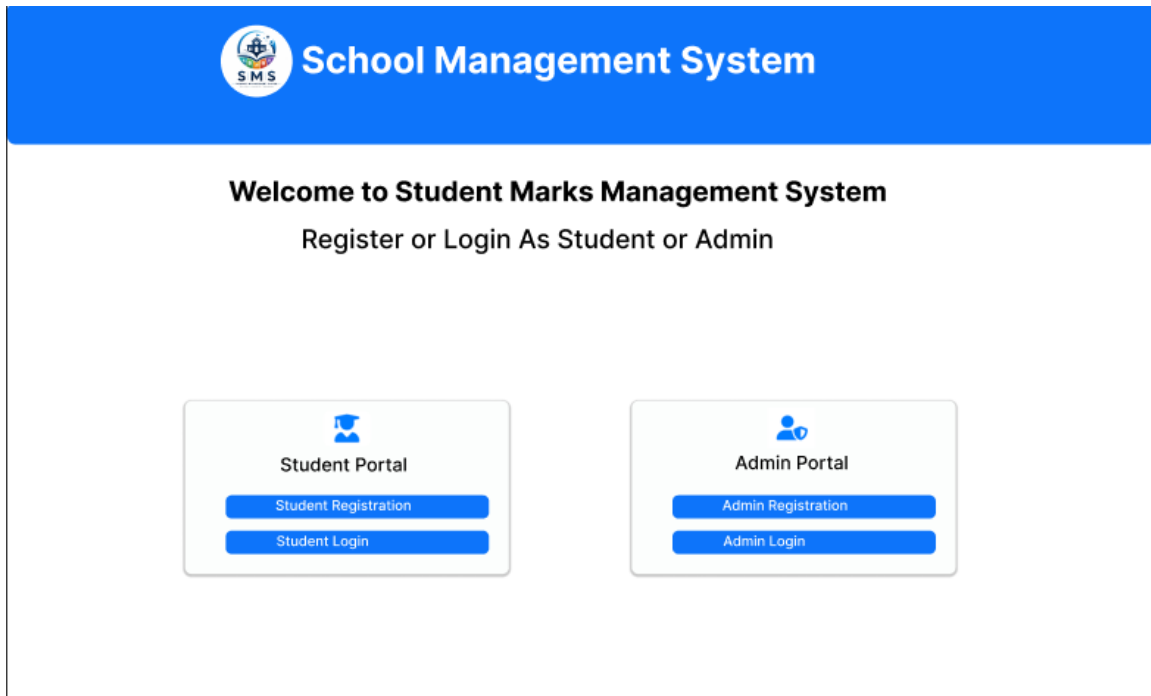


Fig 5: GUI prototype of the School Management System in Figma/Draw.io



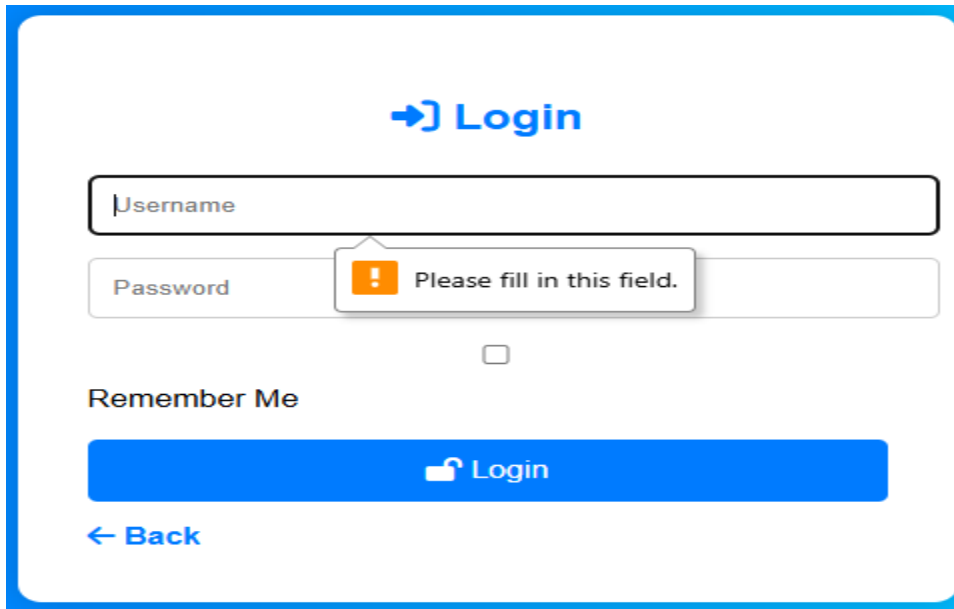
Student reflection

Week 2: Worked on planning and designing interface for Student Management System. Before I implemented it, I created wireframes and dashboard layouts to get a sense of how the system would be structured. The design process taught me to organize user interfaces, plan navigation, colour themes and responsive layouts for desktop and mobile devices.

WEEK 3: JavaScript and PHP Basics
Frontend Interaction and Backend Foundations

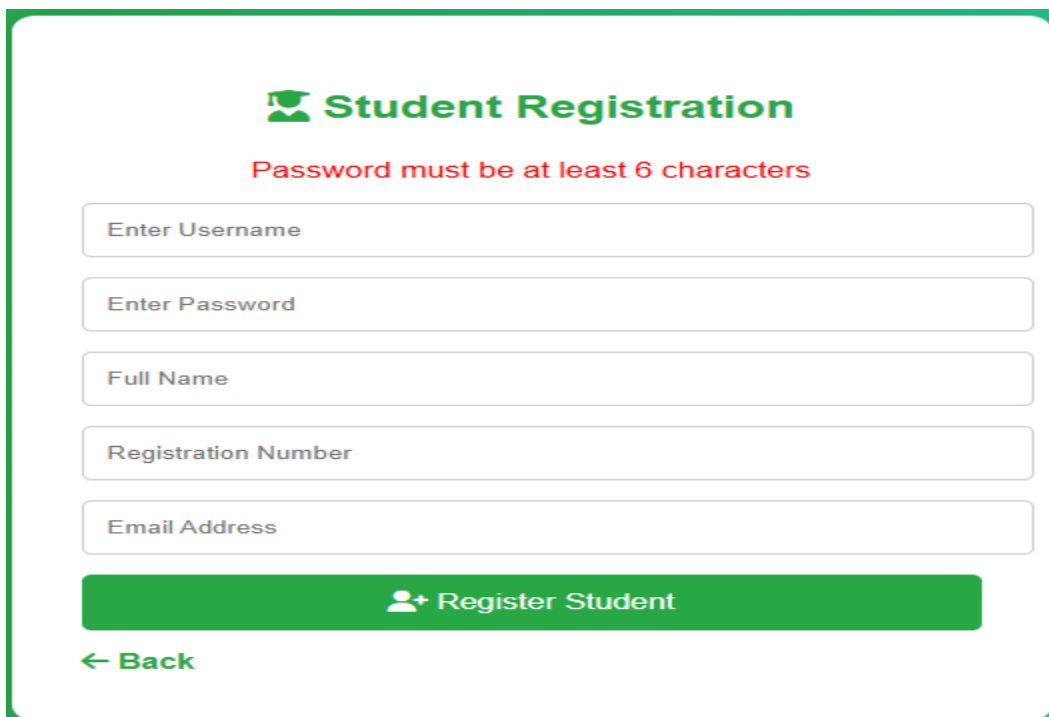
Required Evidence

Fig 1: JavaScript form validation on the login page preventing empty submission



The image shows a login form titled "Login" with a blue arrow icon. It contains two input fields: "Username" and "Password". The "Password" field is currently empty, and a tooltip with an orange exclamation mark icon and the text "Please fill in this field." is displayed above it. Below the password field is a checkbox labeled "Remember Me". At the bottom, there is a blue "Login" button with a white arrow icon and a blue "Back" link.

Fig 2: Password strength checker on the registration form



The image shows a student registration form titled "Student Registration" with a green graduation cap icon. Below the title, a red text message states "Password must be at least 6 characters". The form contains five input fields: "Enter Username", "Enter Password", "Full Name", "Registration Number", and "Email Address". At the bottom, there is a green "Register Student" button with a white plus icon and a green "Back" link.

Fig 3: PHP database connection script (db.php) for MySQL

```
db.php > ...
1  <?php
2
3  /* DATABASE CONFIGURATION */
4
5  $DB_HOST = "localhost";
6  $DB_USER = "root";
7  $DB_PASS = "";
8  $DB_NAME = "school_system";
9
10 /* CREATE CONNECTION */
11
12 $mysqli = new mysqli(
13     $DB_HOST,
14     $DB_USER,
15     $DB_PASS,
16     $DB_NAME
17 );
18
19 /* CHECK CONNECTION */
20
21 if($mysqli->connect_error){
22     die(
23         "Database Connection Failed: "
24         . $mysqli->connect_error
25     );
26 }
27
28 /* SET CHARACTER */
29
30 $mysqli->set_charset("utf8mb4");
31
32 ?>
```

Fig 4: PHP Backend Functions and Database Processing Script

```
function db_prepare_execute($query, $types='', $params=[]){
    global $mysqli;

    $stmt = $mysqli->prepare($query);

    if(!$stmt){
        throw new Exception("Prepare Failed: ".$mysqli->error);
    }

    if(!empty($types) && !empty($params)){
        $stmt->bind_param($types, ...$params);
    }

    if(!$stmt->execute()){
        throw new Exception("Execute Failed: ".$stmt->error);
    }

    return $stmt;
}

function sanitize_string($str){
    return htmlspecialchars(trim($str));
}

function is_logged_in(){
    return isset($_SESSION['user_id']);
}

function is_admin(){
    return isset($_SESSION['role']) &&
           $_SESSION['role'] === 'admin';
}
```

Fig 5: Dynamic user input handling — product search filtering results in real time

Manage Marks - joy wanjiru

Marks

english
mathematics

+ Add Mark

Student Marks

ID	Subject	Marks	Term	Actions	
1	english	80	1	Edit	Delete
2	mathematics	78	1	Edit	Delete

Fig 6: PHP Script Returning JSON Response from the Marks Table

```
{
  {
    "name": "joy wanjiru",
    "reg_no": "reg001",
    "subject": "english",
    "marks": 80,
    "term": "1"
  },
  {
    "name": "joy wanjiru",
    "reg_no": "reg001",
    "subject": "mathematics",
    "marks": 78,
    "term": "1"
  },
  {
    "name": "cate muthoni",
    "reg_no": "reg002",
    "subject": "english",
    "marks": 67,
    "term": "1"
  },
  {
    "name": "cate muthoni",
    "reg_no": "reg002",
    "subject": "mathematics",
    "marks": 77,
    "term": "1"
  },
}
```

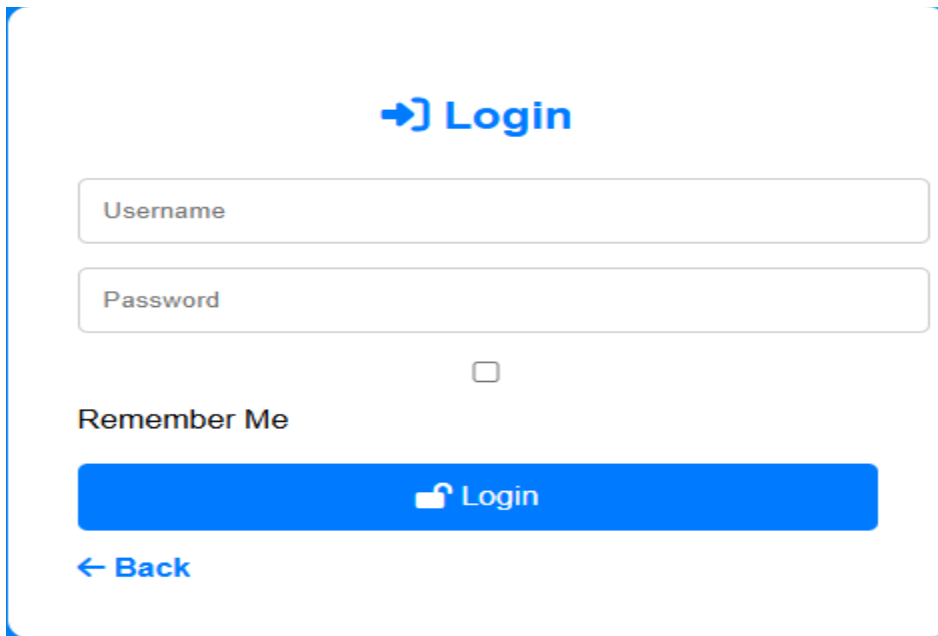
Student Reflection

Week 3 taught me how backend and frontend technologies work together in dynamic web applications. I also worked on Student Management System where I implemented form validation, password security, database connectivity, Json data & dynamic input. This week I learnt more about the PHP programming language, JavaScript validation and back-end processing concepts.

WEEK 4: Backend Development and Authentication

Backend Processing and Authentication Systems

Fig 1: Login Page Authentication Form



A login form with a blue header containing a right-pointing arrow icon and the text "Login". Below the header are two input fields: "Username" and "Password". Under the "Password" field is a small square checkbox. Below the checkbox is the text "Remember Me". At the bottom of the form is a blue button with a white right-pointing arrow icon and the text "Login". To the left of the button is a blue link with a left-pointing arrow icon and the text "Back".

Fig 2: Session-Based Authentication Using PHP Sessions

```
<?php

require_once 'db.php';

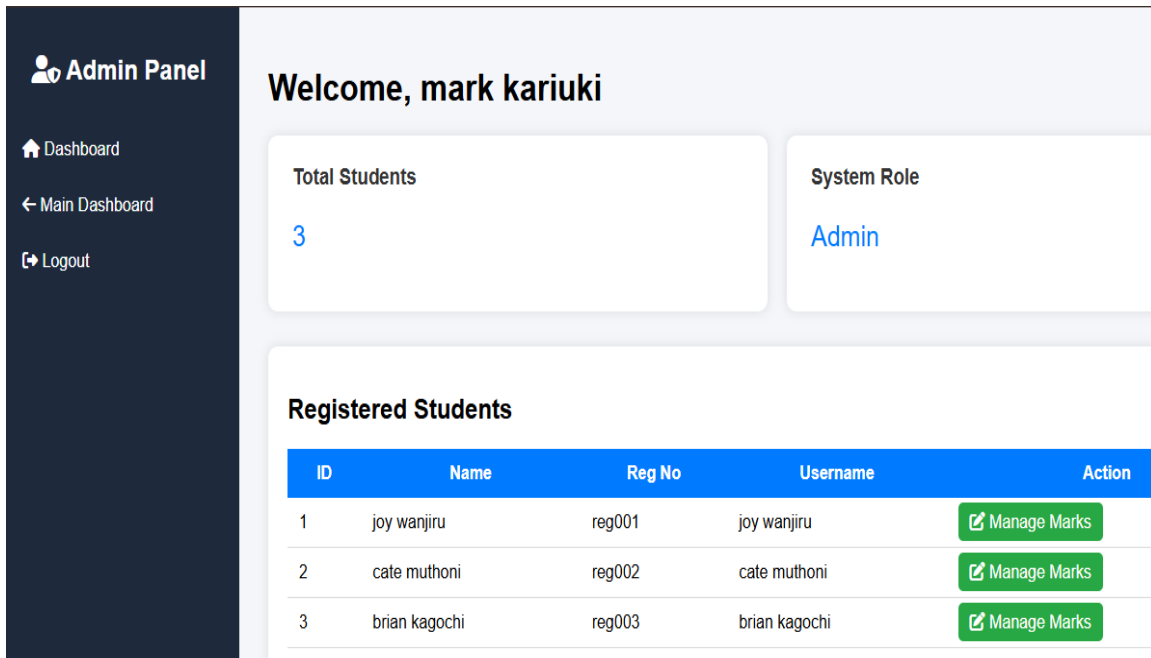
session_start();

if (isset($_SESSION['LAST_ACTIVITY']) &&
    (time() - $_SESSION['LAST_ACTIVITY'] > 1800)) {

    session_unset();
    session_destroy();
}

$_SESSION['LAST_ACTIVITY'] = time();
```

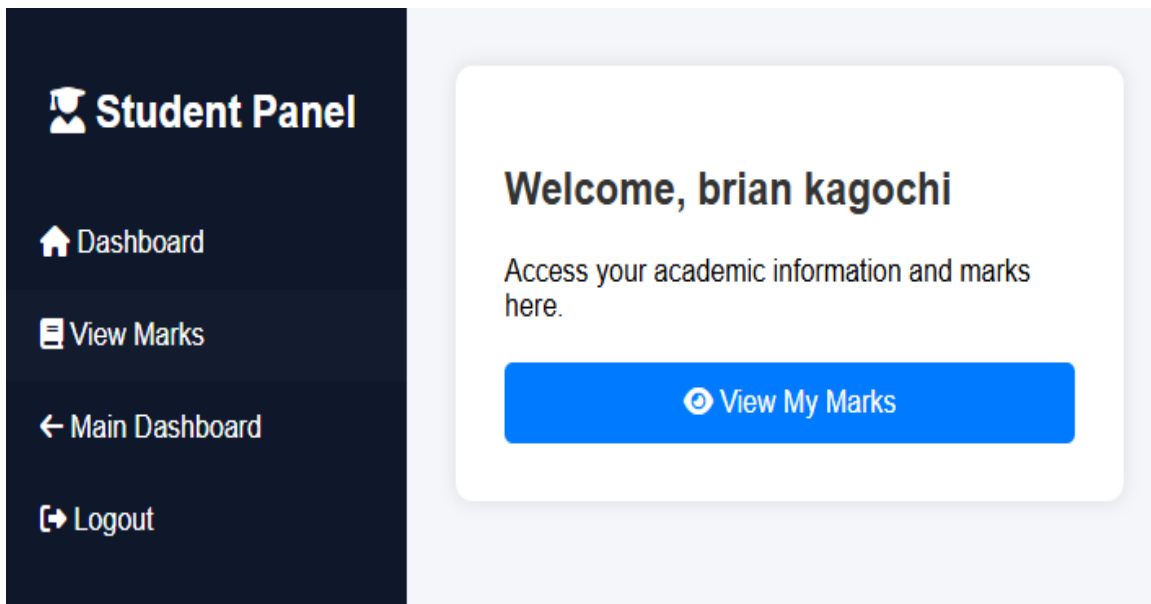
Fig 3: Admin Dashboard After Successful Login



The Admin Dashboard features a dark blue sidebar on the left with the 'Admin Panel' header and navigation links: 'Dashboard' (home icon), 'Main Dashboard' (back arrow), and 'Logout' (logout icon). The main content area is light gray and displays a welcome message 'Welcome, mark kariuki'. Below this, there are two summary cards: 'Total Students' showing '3' and 'System Role' showing 'Admin'. A 'Registered Students' section contains a table with columns for ID, Name, Reg No, Username, and Action. The table lists three students, each with a 'Manage Marks' button.

ID	Name	Reg No	Username	Action
1	joy wanjiru	reg001	joy wanjiru	Manage Marks
2	cate muthoni	reg002	cate muthoni	Manage Marks
3	brian kagochi	reg003	brian kagochi	Manage Marks

Fig 4: Student Dashboard After Successful Login



The Student Dashboard features a dark blue sidebar on the left with the 'Student Panel' header and navigation links: 'Dashboard' (home icon), 'View Marks' (document icon), 'Main Dashboard' (back arrow), and 'Logout' (logout icon). The main content area is light gray and displays a welcome message 'Welcome, brian kagochi'. Below the message, it says 'Access your academic information and marks here.' and features a prominent blue button labeled 'View My Marks' with a circular arrow icon.

Fig 5: Role-Based Authentication Logic in PHP

```
    }

    /* REDIRECT BASED ON ROLE */

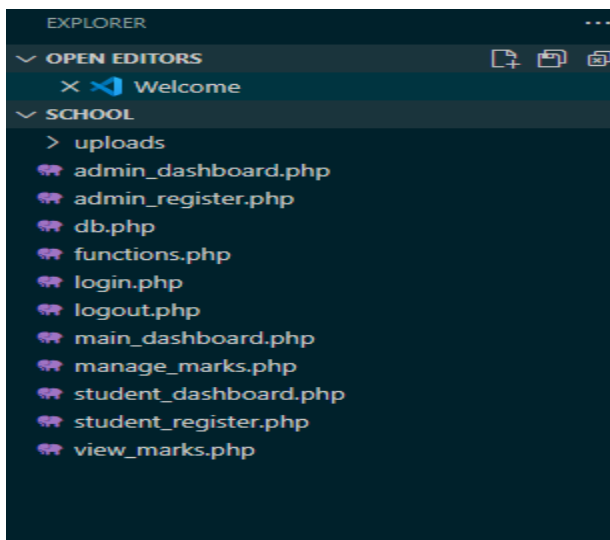
    if($user['role'] === 'admin'){
        header('Location: admin_dashboard.php');
    } else {
        header('Location: student_dashboard.php');
    }

    exit;
}

$err = "Invalid Username or Password";
} catch(Exception $e){
    $err = $e->getMessage();
}
}
?>
```

Fig6: Backend Folder Organization

Professional project structure



Reflection of a Student

In week 4 I learned how backend systems handle user authentication and sessions. Developed a role-based PHP and MySQL login system for students and administrator. This week helped me understand the session management, secure authentication, and backend processing workflows.

WEEK 5: Connecting the Database and CRUD

Database-Driven Application Development

Fig 1: Creating databases using phpMyAdmin.

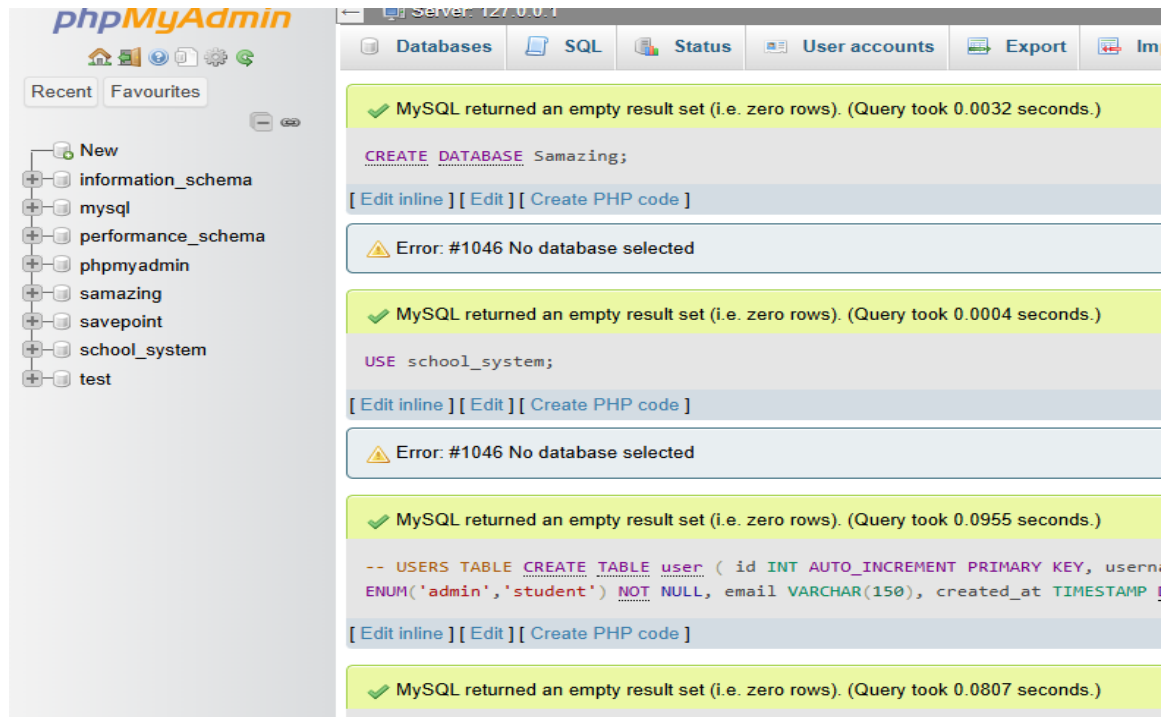
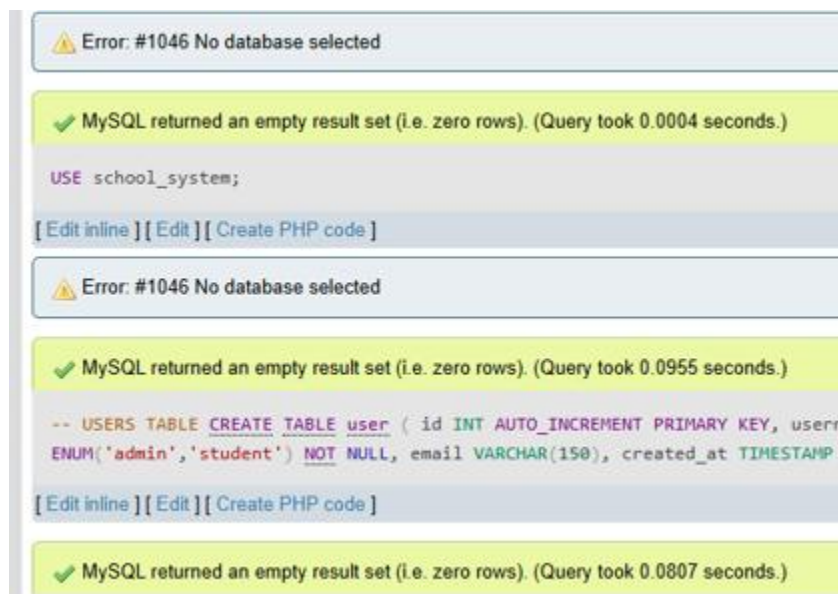


Fig 2: Creating tables inside the database.



CRUD Operations

CRUD = Create, Read, Update, Delete

Fig 3: CREATE (Insert Record)

```
INSERT INTO `marks` (`id`, `student_id`, `subject`, `marks`, `term`, `created_at`) VALUES ('7', '1', 'chemistry', '77', '1', current_timestamp());
```

[Edit inline] [Edit] [Create PHP code]

Show query box

✓ 1 row inserted. (Query took 0.0021 seconds.)

```
INSERT INTO `marks` (`id`, `student_id`, `subject`, `marks`, `term`, `created_at`) VALUES ('8', '1', 'chemistry', '77', '1', current_timestamp());
```

[Edit inline] [Edit] [Create PHP code]

Fig 4: READ (View Records)

✓ Showing rows 0 - 7 (8 total, Query took 0.0028 seconds.)

```
SELECT * FROM `marks`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: I

Extra options

					id	student_id	subject	marks	term	created_at
☐	Edit	Copy	Delete		1	1	english	80	1	2026-05-28 23:19:45
☐	Edit	Copy	Delete		2	1	mathematics	78	1	2026-05-28 23:19:57
☐	Edit	Copy	Delete		3	2	english	67	1	2026-05-28 23:20:35
☐	Edit	Copy	Delete		4	2	mathematics	77	1	2026-05-28 23:20:49
☐	Edit	Copy	Delete		5	3	english	80	1	2026-05-28 23:22:10
☐	Edit	Copy	Delete		6	3	mathematics	77	1	2026-05-28 23:22:21
☐	Edit	Copy	Delete		7	1	chemistry	77	1	2026-06-03 12:57:55
☐	Edit	Copy	Delete		8	1	chemistry	77	1	2026-06-03 12:58:45

↑ ☐ Check all With selected: Edit Copy Delete Export

Fig 5: UPDATE (Edit Record)

✓ 1 row affected.

```
UPDATE `marks` SET `marks` = '78' WHERE `marks`.`id` = 8;
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 7 (8 total, Query took 0.0010 seconds.)

```
SELECT * FROM `marks`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	student_id	subject	marks	term	created_at			
☐		Edit		Copy		Delete	1	1	english	80	1	2026-05-28 23:19:45
☐		Edit		Copy		Delete	2	1	mathematics	78	1	2026-05-28 23:19:57
☐		Edit		Copy		Delete	3	2	english	67	1	2026-05-28 23:20:35
☐		Edit		Copy		Delete	4	2	mathematics	77	1	2026-05-28 23:20:49
☐		Edit		Copy		Delete	5	3	english	80	1	2026-05-28 23:22:10
☐		Edit		Copy		Delete	6	3	mathematics	77	1	2026-05-28 23:22:21
☐		Edit		Copy		Delete	7	1	chemistry	77	1	2026-06-03 12:57:55
☐		Edit		Copy		Delete	8	1	chemistry	78	1	2026-06-03 12:58:45

Fig 6: DELETE (Delete Record)

```
SELECT * FROM `marks`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | ✓ 1 row deleted. (Query took 0.0034 seconds.) | key: N

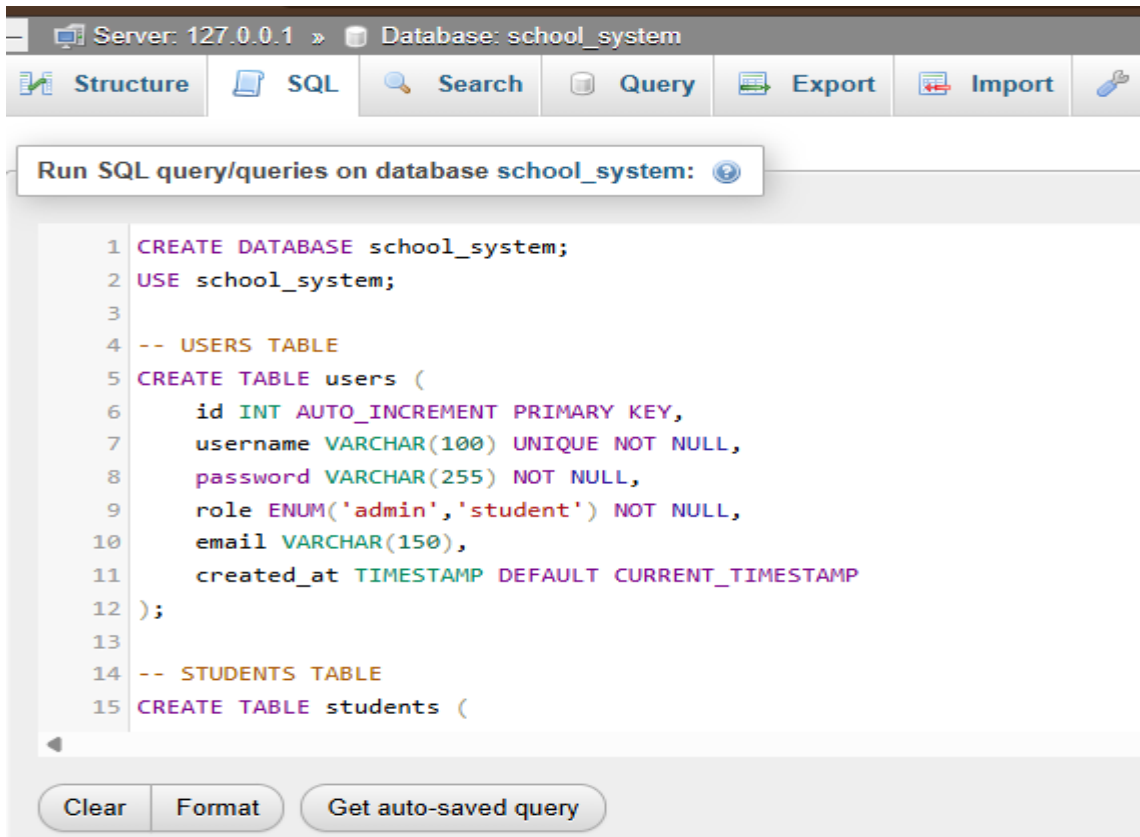
Extra options

				id	student_id	subject	marks	term	created_at			
☐		Edit		Copy		Delete	1	1	english	80	1	2026-05-28 23:19:45
☐		Edit		Copy		Delete	2	1	mathematics	78	1	2026-05-28 23:19:57
☐		Edit		Copy		Delete	3	2	english	67	1	2026-05-28 23:20:35
☐		Edit		Copy		Delete	4	2	mathematics	77	1	2026-05-28 23:20:49
☐		Edit		Copy		Delete	5	3	english	80	1	2026-05-28 23:22:10
☐		Edit		Copy		Delete	6	3	mathematics	77	1	2026-05-28 23:22:21

☐ Check all
 With selected:
 Edit
 Copy
 Delete
 Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: N

Fig 7: SQL Database Table Creation Script



Reflection of a Student

Week 5 was about learning how to connect PHP applications to MySQL databases and how to do CRUD operations. I added a feature to manage marks, where an admin can dynamically add, edit, update, and delete records. This week's learning has increased my understanding of database driven systems and backend web application development.

References

1. PHP Documentation – <https://www.php.net>
2. MySQL Documentation – <https://dev.mysql.com/doc>
3. XAMPP Documentation – <https://www.apachefriends.org>
4. W3Schools – <https://www.w3schools.com>
5. MDN Web Docs – <https://developer.mozilla.org>
6. GitHub Documentation – <https://docs.github.com>